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Article

Quantum Machine Learning and Data Re-Uploading: Evaluation on Benchmark and Laboratory Medicine Data Sets

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BACKGROUND: Quantum machine learning (QML) is an emerging field that may offer unique advantages over classical machine learning but has not been extensively studied with real-world healthcare data of practical size. This study evaluates the performance of a recently proposed quantum machine learning algorithm—quantum circuits with data re-uploading (QC-REUP)—in comparison to classical machine learning and other QML methods. Performance was assessed using both benchmarking data sets and real-world laboratory medicine data.

METHODS: Four data sets containing between 2 and 30 features were selected for evaluation. Initial baseline comparisons of classification performance (F1 score) were conducted using all algorithms—QC-REUP, 2 QML, and 4 classical machine learning (ML) algorithms—across the four data sets. Configuration parameters were then optimized for the QC-REUP algorithm using a previously published data set of plasma amino acid (PAA) profiles to determine the impact of optimization on classification performance, followed by a final comparison against classical ML algorithms.

RESULTS: Baseline comparisons showed that QC-REUP outperformed quantum and linear classical algorithms on lower-dimensional data sets. However, as input dimensionality increased, QML F -score declined. Following optimization, QC-REUP improved relative to baseline, and again performed comparably to linear algorithms, but ultimately demonstrated lower performance than nonlinear classical ML algorithms on the PAA data set.

CONCLUSION: This study suggests that data re-uploading algorithms can perform comparably to classical approaches in specific contexts, particularly with low-dimensional data. While optimization can enhance performance, further improvements in quantum hardware and algorithmic development are likely needed before QML can be effectively applied in laboratory medicine and broader biomedical research.

Introduction

In the era of personalized medicine, there is growing interest in using data analytics to deliver more tailored diagnoses and treatments. Today, artificial intelligence (AI) and machine learning (ML) play important roles in these efforts and have been widely implemented in various clinical specialties, including laboratory medicine (1–3). ML has been successfully applied to diverse tasks in the clinical laboratory, such as image classification, quality assurance, and disease risk prediction (2, 4). However, current ML algorithms face challenges in maintaining interpretability, generalizing from smaller data sets, and ensuring equitable predictive performance across subgroups (5–7). Consequently, there is an ongoing need to evaluate novel ML algorithms and explore solutions that may enhance the capabilities of ML applications in healthcare and laboratory medicine.

Quantum computing is an emerging field that offers theoretical advantages over classical computing (8, 9). By leveraging basic principles of quantum mechanics, quantum computers (QCs) can represent multiple computational states simultaneously (9). This capacity allows QCs to theoretically perform calculations more efficiently than classical computers (10, 11). Over the last decade, access to QCs has increased with many available through public cloud platforms with vendors offering ongoing improvements in quantum processing unit (QPU) size and error mitigation strategies (8). In addition, the development of

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open-source quantum software development kits (QSDKs) and QPU simulators have allowed for increased adoption across many scientific domains. As a result, there has been a growing number of publications describing novel quantum algorithms and investigating the potential use of QCs for practical applications, using both QPU simulators and actual QPUs (8).

Quantum machine learning (QML) is a subfield of quantum computing that both adapts classical ML methods to quantum frameworks and introduces quantum-native algorithms that do not have direct classical counterparts (8, 12). By leveraging quantum properties such as superposition and entanglement, QML may offer benefits such as reduced training time, improved accuracy, and better performance on small imbalanced data sets (13, 14). Recent research has investigated the utility of QML for common tasks like classification, regression, clustering, and feature selection (15–17). While some studies show promising classification performance on low-dimensional data sets (i.e., only a few independent variables), relatively little work considers non-synthetic, higher-dimensional data sets (i.e., more independent variables—e.g., metabolomic profile), such as those commonly encountered in real-world applications. This is understandable, as increasing the number of qubits in a quantum circuit makes the system more susceptible to noise and decoherence, which can introduce errors during computation (9, 15, 18). This issue is particularly limiting for some QML algorithms, such as quantum neural networks (QNNs), which require larger quantum circuits (i.e., more qubits) to accommodate more input features (8, 19). Quantum simulators are also insufficient in these situations because they require a large amount of memory to simulate these models on classical hardware. To overcome these challenges, novel approaches to quantum algorithm and circuit design have recently been proposed.

In 2020, Pérez-Salinas et al. described a new algorithm for QML classification tasks that requires minimal quantum resources and smaller quantum circuits relative to other QML algorithms, such as QNNs (20). They showed that a single qubit, combined with data re-uploading, can function as a universal classifier and may enable the approximation of continuous functions (20). With fewer qubits required, this approach offers a valuable opportunity to evaluate QML classification performance on practical, higher-dimensional data sets while still leveraging quantum properties. Subsequent studies have evaluated data re-uploading but with a primary focus on theoretical foundations, lower-dimensional synthetic data sets, or higher-dimensional data sets with feature reduction (21–24). While these works provide valuable insights, there remains a need to assess the performance of data re-uploading algorithms with non-synthetic, clinical data sets.

While QML algorithms may offer efficiency gains, their value ultimately depends on achieving acceptable baseline accuracy. This study evaluated the performance of quantum circuits with data re-uploading (QC-REUP) compared to state-of-the-art classical and quantum ML algorithms on benchmark data sets. We also systematically examined different configurations of QC-REUP, evaluating the effects of circuit size, qubit entanglement, and data re-uploading strategies on classification performance. To assess the short-term potential of QML in healthcare, we applied these methods to a published clinical data set from Wilkes et al., containing 28 input features from plasma amino acid (PAA) profiles obtained during routine clinical testing for inherited metabolic diseases (25). The evaluation of the PAA data set represents a practical biomedical QML application that has been benchmarked using classical ML methods (25, 26).

Materials and Methods

TECHNOLOGY

All experiments used Python (version 3.10) within version-controlled, virtualized Miniconda environments. Quantum-based algorithms were implemented using the Qiskit SDK (version 1.0.2; IBM). A grid search of all configuration parameters was conducted and logged using the Azure Machine Learning/mlflow library (version 2.12.2). All analyses were run on Azure serverless virtual machines equipped with traditional CPUs (48 cores, 96 GB RAM, 384 GB disk). All QML algorithms in this study were executed using quantum simulators on the same serverless virtual machines. Classical ML algorithms used scikit-learn (version 1.4.2), xgboost (version 2.0.3), and tensorflow (version 2.16.1). Feature scaling methods were implemented for continuous variables using the scikit-learn package (version 1.4.2). *t*-Tests were performed with the python package SciPy (version 1.13.0).

DATA SET SELECTION

We evaluated the performance of classical and quantum ML algorithms using 4 data sets (Fig. 1). These data sets were selected to represent both low- and high-dimensional input feature spaces and are well documented in the literature regarding classical ML performance. Three of the data sets are commonly used for ML benchmarking: the (*a*) Circle, (*b*) Iris, and (*c*) Wisconsin Diagnostic Breast Cancer (WDBC) data sets (27, 28).

The Circle data set is a synthetic, two-dimensional data set with a binary classification where the classes are separated by a single circular boundary (Supplemental Fig. 1). The Iris data set is widely used in ML and contains 150 samples from 3 species of iris flowers, with 4 features (sepal length, sepal width, petal length, and

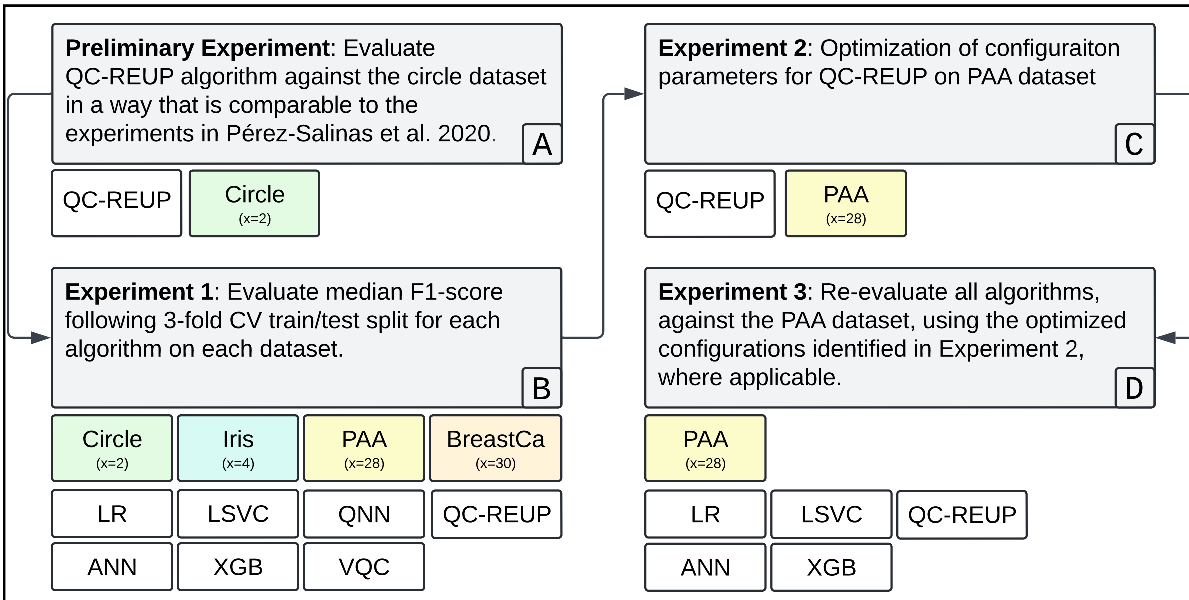

Fig. 1. Overview of study design and experimental workflow: (A), baseline evaluation of the QC-REUP implementation using the Circle data set; (B), comparison of classical ML and QML algorithms across all data sets; (C), parameter optimization for QC-REUP using the PAA data set; and (D), final comparison of optimized QC-REUP algorithm with classical ML algorithms on the PAA data set. Color figure available at clinchem.org.

petal width). For the Iris data set, two of the three mutually exclusive classes were combined into one class to convert the data into a binary classification problem. The WDBC data set consists of 569 instances with 30 features derived from digitized images of fine-needle aspirate samples of breast masses with a binary classification of benign or malignant. The fourth data set, the PAA data set, was obtained from routine clinical chemistry testing as described by Wilkes et al. It contains 2084 anonymized PAA profiles, each with 28 features, and is enriched for rare disease cases (25). Each record is labeled as either normal or abnormal.

ALGORITHM SELECTION

For comparison against QC-REUP, we selected 4 classical ML algorithms for this study, with the intent to include those which are capable of learning linear and nonlinear decision boundaries. Specifically, we evaluated the performance of logistic regression, a linear support vector classifier (LSVC), extreme gradient boosting (XGB), and an artificial neural network (ANN). For QML algorithms, we implemented a variational quantum classifier (VQC) and a quantum neural network (QNN).

In this study, a custom implementation of the QC-REUP algorithm, as outlined by Pérez-Salinas et al., was developed by the authors, using the Qiskit library and implemented in Python. The QC-REUP algorithm followed the architecture described by Pérez-Salinas et al. and used a single qubit and 2 data reupload layers for the initial baseline comparison between classical and quantum ML algorithms (Fig. 2) (20). Configurations for the QNN algorithm were chosen based on configuration parameter optimization performed in a prior study by the current authors (29). The VQC algorithm used an "out-of-the-box" configuration. The ANN algorithm architecture was a fully connected, feed-forward neural network with an input layer, followed by a single hidden layer containing 64 neurons activated by the rectified linear unit (ReLU) function. The output layer was a single neuron with a sigmoid activation function, and the network was compiled for binary classification using the Adam optimizer, binary cross-entropy loss, and accuracy as the performance metric. All other algorithms were implemented using default settings unless otherwise described.

EXPERIMENTAL DESIGN: CLASSICAL AND QML COMPARISON

The overall study design and experiment flow can be visualized in Fig. 1. Unless otherwise noted, all experiments used a 70:30 train-test split and 3-fold cross-validation to assess performance variability on the test data set. All experiments use the F1 score, or additional metrics if specified, and were calculated from predictions on the test data set. The F1 score, the harmonic mean of sensitivity and positive predictive

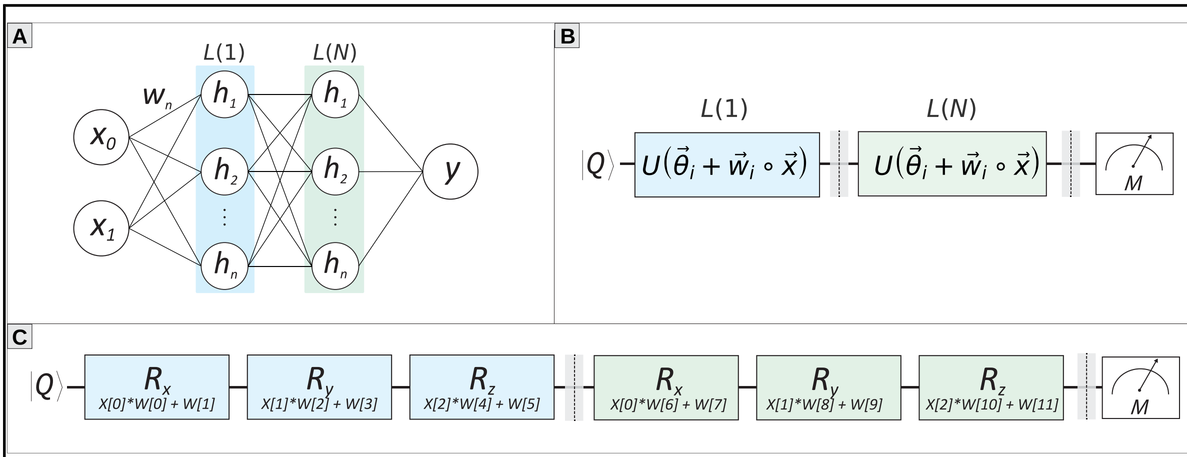

Fig. 2. (A), Diagram of a fully connected, feed-forward artificial neural network where input features (*x*) are received as weighted inputs (*w*) into neurons (*h*) arranged in layers (*L*), where L_1 is shaded blue, and L_n is shaded green. Each neuron (box) applies a transformation (e.g., weighted sum and activation function) and passes the result to the next layer; (B), Quantum circuit diagram for data re-uploading with a single qubit register (Q). Classical input features are encoded via parameterized unitary transformations ($U(\Theta_i + w_i \circ x)$), arranged in layers (*L*), and measured (M). (C), Explicit encoding of features (x_i) into quantum gates (R_x, R_y, and R_z), applied to qubit (Q), with their respective rotational planes (e.g., X-, Y-, or Z-plane). Input (X) and weight (W) arrays with the associated indices, indicate where features and weights are applied. Dashed lines mark boundaries of "re-upload layers." Taken together, this figure illustrates a quantum circuit with a single qubit and 2 data re-upload layers, as it compares to a conceptually analogous ANN architecture. Color figure available at clinchem.org.

value, is a continuous metric ranging from 0 to 100, with 100 indicating perfect classification.

To verify the correctness of the author's QC-REUP implementation, the algorithm was first tested on the Circle data set following a similar approach to that described by Pérez-Salinas et al. (Figure 1A). Classical ML and QML algorithms were then evaluated on each data set and compared by F1 score (Fig. 1B). Following this baseline comparison, optimization of configuration parameters was performed for the QC-REUP algorithm using the PAA data set (Fig. 1C). Once the optimal configurations were determined, the QC-REUP was again compared to classical ML algorithms, this time only using the PAA data set (Fig. 1D). To provide a more balanced comparison with the classical ANN algorithm, layer architecture was adjusted to approximately match the number of trainable parameters in the QC-REUP algorithm. A paired one-tailed *t*-test was conducted to compare the F1 score of the QC-REUP algorithm against the classical ML algorithms, across 5-fold cross-validation trials to assess whether classical algorithms significantly outperformed QC-REUP using optimized configurations. Statistical significance by *t*-test was evaluated at 0.0125 after Bonferroni correction.

QUANTUM CIRCUIT DATA RE-UPLOADING CONFIGURATION PARAMETER OPTIMIZATION

In this study, configuration parameters refer to settings established before training that can affect the model's performance. These parameters include those related to data set preparation (e.g., feature scaling), quantum circuit design, and the method by which data is uploaded to the quantum circuit (Table 1). Data re-uploading methods include symmetrical and asymmetrical approaches and are applicable only when there is more than one qubit. These methods determine how input features are distributed across multiple qubits. For instance, in a quantum circuit with more than one qubit, a symmetrical re-upload approach applies each input feature (x_i) to all qubits. In contrast, an asymmetrical approach sequentially uploads input features to qubit 1 and then to qubit 2, ensuring equal distribution across the available qubits.

To evaluate the impact of each configuration parameter on QC-REUP classification performance by F1 score, we analyzed Shapley additive explanations (SHAP) values and feature importance by random forest regression. In addition, we examined the frequency of each parameter value's occurrence within the top 10% of grid-search runs, as ranked by F1 score.

Table 1. Options for quantum circuit data re-upload configuration parameter grid-search.

Configuration parameter	Options
Downsample negative class	True, False
Feature scaler	Standard, min0max1
Number of qubits	1, 2, 4
Number of re-upload layers	2, 4, 6, 8
Quantum entanglement	True, False
Quantum re-upload method	Symmetrical, Asymmetrical
Quantum noise	True, False

Table 2. Median F1 score across 3-fold cross-validation with associated 95% confidence interval.

	Data Set[a]			
Algorithm	Circle (2)	Iris (4)	PAA (28)	WDBC (30)
Linear regression	0.51 (0.51–0.54)	0.68 (0.60–0.74)	0.62 (0.60–0.62)	0.91 (0.91–0.96)
Linear SVC	0.51 (0.51–0.54)	0.72 (0.58–0.75)	0.45 (0.45–0.62)	0.87 (0.81–0.91)
Neural network	0.98 (0.98–0.99)	0.87 (0.83–0.93)	0.70 (0.70–0.74)	0.92 (0.91–0.96)
XGBoost	0.98 (0.97–0.98)	0.90 (0.89–0.96)	0.85 (0.84–0.85)	0.96 (0.92–0.98)
QC-REUP	0.90 (0.88–0.91)	0.86 (0.83–0.96)	**0.40 (0.36–0.41)**[b]	0.44 (0.41–0.45)
QNN	0.85 (0.85–0.87)	0.63 (0.37–0.71)	DNF	DNF
VQC	0.86 (0.86–0.87)	0.48 (0.48–0.71)	DNF	DNF

[a]Number of input features are indicated in parentheses next to the data set.
[b]The performance of the algorithm of interest, QC-REUP, on the PAA data set is highlighted in bold.
Abbreviations: DNF, did not finish; SVC, support vector classifier; XGBoost, extreme gradient boosting; QNN, quantum neural network; VQC, variational quantum classifier.

DATA AND CODE AVAILABILITY

The Circle data set was generated as described in the previous section and is available as part of the project's code. The Iris and WDBC benchmark ML data sets are publicly available from various sources (30). PAA data are available from the original publication (25). The code was independently written and reviewed by 2 authors and is available at https://github.com/acomphealth/quantum–reup. A static version can also be found at https://doi.org/10.5281/zenodo.18249315.

Results

INITIAL CLASSICAL AND QML COMPARISON

The QC-REUP algorithm we implemented demonstrated comparable performance to the originally described implementation on the circle data set (Supplemental Table 1). For the initial baseline comparison between classical ML and QML models (Fig. 1B), the QC-REUP model demonstrated comparable performance to the classical ML algorithms on lower-dimensional data sets, with QC-REUP demonstrating median F1 scores of 0.90 (95% CI, 0.88–0.91) and 0.86 (95% CI, 0.83–0.96) on the Circle and Iris data sets, respectively. However, the QC-REUP model demonstrated a significant decline in classification performance with higher-dimensional input feature spaces with median F1 scores of 0.40 (95% CI, 0.36–0.41) and 0.44 (95% CI, 0.41–0.45) on the PAA and WDBC data sets, respectively. Review of the QC-REUP training loss values demonstrated a failure to minimize on the PAA data set (Supplemental Fig. 2). Some algorithms, such as the QNN and VQC, failed to converge or exceeded computational resources and therefore did not finish training on higher-dimensional data sets (Table 2).

QUANTUM CIRCUIT DATA RE-UPLOADING CONFIGURATION PARAMETER OPTIMIZATION

Using the predefined configuration parameter search space (Table 1), approximately 200 unique runs were

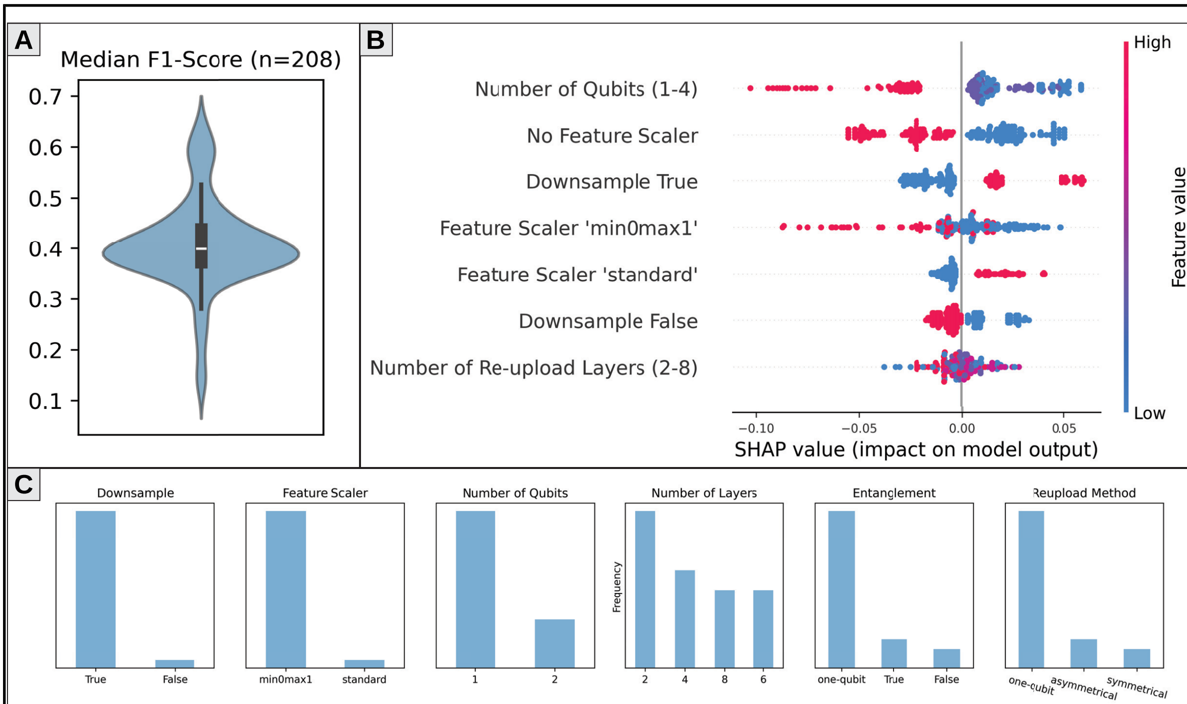

Fig. 3. (A), Violin plot of median F1 score for QC-REUP algorithm on PAA data set for all trials that completed successfully; (B), SHAP summary plot for quantum circuit data re-uploading configuration parameter optimization. Binary features are one-hot encoded, and features with minimal or no contribution are omitted; (C), Frequency of configuration parameters among the top 10% of sweep trials based on median F1 score. Color figure available at clinchem.org.

conducted with the QC-REUP algorithm on the PAA data set (Fig. 1C). Runs using a noisy quantum simulator were excluded from the final analysis due to these requiring excessive computational resources and computation time (>48 h). The median F1 scores across all runs ranged from 0.13 to 0.64, with a median of 0.40 (95% CI, 0.19–0.61) (Fig. 3A). SHAP values indicated that downsampling the negative class, feature scaling, and the number of qubits were the most influential factors on F1 score (Fig. 3B). This was also corroborated by the feature importance analysis by random forest regression (Supplemental Table 2). The frequency of parameters in the top 10th percentile of F1 scores aligned with the SHAP analysis; however, showed a slight advantage for feature scaling (range 0 to 1) (Fig. 3C). In addition, increasing the number of qubits generally had a negative impact on F1 scores, while increasing the number of layers had a mixed impact.

OPTIMIZED QUANTUM CIRCUIT DATA RE-UPLOADING COMPARISON ON PAA DATA SET

The comparison of the classical ML algorithms to the QC-REUP, using the PAA data set, used the following configuration parameters based on our optimization results: negative class downsampling, data scaling (range 0 to 1), one qubit, and 6 data-reupload layers (Supplemental Fig. 3). Examination of the QC-REUP train loss values demonstrated an appreciable decrease over training iterations, indicating optimization convergence (Supplemental Fig. 4). The QC-REUP algorithm demonstrated a median classification accuracy of 0.74 (95% CI, 0.64–0.79) and an improved median F1 score of 0.61 (95% CI, 0.48–0.64). The difference in classification performance between QC-REUP and logistic regression or LSVC was not statistically significant. However, classification performance of the ANN and XGB demonstrated significantly higher classification performance based on F1 score (Table 3).

Discussion

In this study, we demonstrate that QC-REUP can achieve classification performance comparable to classical ML models on lower-dimensional data sets (i.e., fewer independent variables), consistent with prior studies (20, 31–33). Following optimization, QC-REUP can

Table 3. Median and 95th percentile for specified performance metrics on the PAA data set across 5-fold cross validation.

Algorithm[a]	Accuracy	Recall	Precision	F1 Score
Linear Regression	0.77 (0.75–0.79)	0.65 (0.59–0.67)	0.61 (0.59–0.67)	0.64 (0.59–0.66)
Linear SVC	0.79 (0.78–0.81)	0.65 (0.60–0.69)	0.66 (0.64–0.70)	0.65 (0.63–0.70)
Neural Network	0.86 (0.85–0.90)	0.70 (0.66–0.81)	0.81 (0.80–0.86)	0.75 (0.73–0.83)[b]
XGBoost	0.91 (0.90–0.93)	0.90 (0.88–0.93)	0.83 (0.78–0.87)	0.86 (0.85–0.89)[b]
QC-REUP	0.74 (0.64–0.79)	0.60 (0.53–0.71)	0.58 (0.43–0.67)	**0.61 (0.48–0.64)**[c]

[a]Abbreviations: SVC, support vector classifier; XGBoost, extreme gradient boosting; QC-REUP, quantum circuits with data re-uploading.
[b]*P* value <0.0125 as compared to QC-REUP median F1 score classification performance.
[c]The F1 score of the algorithm of interest, QC-REUP, is highlighted in bold.

be successfully trained on higher-dimensional data sets, without the need for feature reduction, and achieve classification performance similar to classical linear ML algorithms. However, performance declines were observed when applying QML, including data reuploading, to more complex and higher-dimensional feature spaces (i.e., more independent variables), which are common in healthcare and biomedical data sets.

In our initial evaluation (Fig. 1B), the QNN and VQC algorithms demonstrated reduced classification performance as the number of input features increased from 2 (Circle data set) to 4 (Iris data set), and failed to train on larger data sets (PAA and WDBC) due to convergence issues and excessive runtimes. Training failures on the larger data sets for QNN and VQC was somewhat expected given their one-to-one feature-to-qubit mapping, which requires wider quantum circuits (i.e., more qubits) and substantially more computational resources for quantum simulation (8). In contrast, QC-REUP achieved higher F1 scores than other QML algorithms on the Circle and Iris data sets but declined significantly on the higher-dimensional WDBC and PAA data sets. The cause of this decline as the input feature space increased from 4 to 30 remains unclear but may indicate a limitation of the re-uploading approach in learning nonlinear decision boundaries in higher-dimensional spaces.

Optimization of configuration parameters for QC-REUP on the PAA data set (Fig. 1C) led to improvement in classification performance but also highlighted challenges in tuning QML algorithms. The wide variation in F1 scores (0.12 to 0.64) underscores the algorithm's sensitivity to configuration parameter choices. Feature scaling and downsampling the negative class during data preprocessing were among the most influential factors affecting classification performance, suggesting that QC-REUP is sensitive to input feature range and class imbalance. Additionally, increasing the number of qubits generally decreased classification performance, while increasing the number of layers had a more variable effect on F1 scores. When using an ideal quantum simulator—where quantum noise and decoherence do not contribute to information loss—this was an unexpected result, as more trainable parameters would typically be expected to improve model performance (34, 35).

In classical ML, classification performance can be improved by tuning configuration parameters during the training process (36). While this can lead to notable performance improvement on test sets, the variation reported in most classical ML literature is not as pronounced as what we observed in our QML experiments (22, 37, 38). Of note, a similar degree of variation and sensitivity in classification performance to parameter choices was observed in our previous work with QNNs (29). Although a direct comparison with classical ML parameter optimization was outside the scope of this study, it should be acknowledged that the exact degree of performance variability likely depends on the specific data set, model, and experimental setup. Nevertheless, these observations underscore the importance of incorporating similar evaluation and optimization protocols in future QML research.

Despite the initial challenges encountered with higher-dimensional data sets, optimization of configuration parameters for the QC-REUP algorithm significantly improved classification performance on the PAA data set, with an increase in median F1 score from 0.40 (95% CI, 0.36–0.41) to 0.61 (95% CI, 0.48–0.64). This indicates that optimized quantum classification circuits with data re-uploading can be successfully trained on higher-dimensional biomedical data sets, using currently available quantum simulation, without feature reduction, while other QML algorithms cannot. When considering classification performance relative to classical ML algorithms (Fig. 1D), the median F1 score of the optimized QC-REUP algorithm was not significantly different compared to linear classical ML algorithms (Table 3).

For comparison with the classical ANN, the network architecture was adjusted to approximate the number of trainable parameters in the optimized QC-REUP algorithm (approximately 360). Despite this adjustment, the ANN achieved significantly higher classification performance with a median F1 score of 0.75 (95% CI, 0.73–0.83). Potential contributing factors to this performance gap include challenging optimization landscapes, where gradients vanish or get trapped in poor local minima, making it difficult for the model to converge. Additionally, practical constraints on qubits, layers and parameterization may limit their expressivity, restricting how complex a decision boundary they can model for high-dimensional data. While QC-REUP algorithms are theoretically universal function approximators, the observed performance gap underscores the need for further research in algorithm design and evaluation with higher-dimensional data sets to further clarify the practical limits of their expressive capability (21, 31).

A key objective of this study was to evaluate the QC-REUP algorithm against classical ML counterparts, using data sets with dimensionality and complexity typical of real-world healthcare applications. QML algorithms leveraging quantum properties, such as superposition and entanglement, are hypothesized to offer advantages, including faster training times, improved accuracy, and better performance on smaller imbalance data sets (8, 13, 39). To this end, the evaluation of QML performance is highly relevant to the fields of academic medicine and laboratory medicine. However, while these potential advantages are promising, they remain theoretical until QML algorithms can consistently match or exceed the performance of well-established classical counterparts—a standard not yet consistently achieved (39).

This study had several limitations. First, we were unable to assess the impact of noise on QML classification performance due to prolonged run time and computational errors. Quantum noise refers to unintended disturbance of quantum states, like superposition and entanglement, which can degrade information and cause computational inaccuracy (8). While the simulators used in this study can model noise, even noiseless runs from Experiment-2 averaged 13.5 (SD 11.8) h, with some exceeding 60 h, underscoring the resource demands of simulating QML on classical hardware without feature reduction.

Second, all QML algorithms were tested on quantum simulators rather than QPUs. Although QPUs are increasingly accessible via cloud platforms, we did not include them due to the need for additional optimization and high cost. IBM Quantum® currently charges $96 per minute for QPU access, including associated classical computation time. If our simulation run times approximated QPU run time, Experiment-2 alone would have cost over $18 million. Despite this limitation, our results provide meaningful insight, showing that QC-REUP circuit architectures can successfully model high-dimensional data sets effectively without the need for feature reduction. Whether performance on QPUs would be better or worse remains uncertain, as real quantum effects may enhance or complicate learning. However, in related work by the authors, noise modeling led to a mild performance drop (29). Lastly, while the PAA data set has higher dimensionality, it may not reflect the full complexity of clinical data sets, limiting the generalizability of our findings on learning from large input spaces.

Overall, these findings show that QC-REUP can outperform classical linear and QML algorithms on lower-dimensional data sets, can be trained on higher-dimensional data sets without feature reduction, and can be optimized to perform comparably to classical linear models on real-world, benchmarked data. However, classical nonlinear algorithms such as XGBoost and ANNs consistently performed better. We also observed substantial performance gains through optimization, highlighting the importance of tuning configuration parameters in QML studies.

As quantum hardware and algorithms continue to evolve, QML may overcome current limitations. Ongoing evaluation remains important in healthcare and laboratory medicine. Future research should explore hybrid quantum–classical models, data-driven optimization strategies, and feature reduction methods—such as principal component analysis or recursive feature elimination—to support QML in higher-dimensional settings. In the near term, QML may complement classical ML for lower-dimensional or feature-selected data sets. In the longer term, QML could play a broader role in ML applications, though significant technological and methodological advances will likely be necessary to realize its full potential.

Supplemental Material

Supplemental material is available at *Clinical Chemistry* online.

Author Declaration: A version of this paper was previously posted as a preprint on medRxiv as https://doi.org/10.1101/2025.05.14.25327605.

Nonstandard Abbreviations: QML, quantum machine learning; QC-REUP, quantum circuits with data re-uploading; ML, machine learning; PAA, plasma amino acid; AI, artificial intelligence; QC, quantum computer; QSDK, quantum software development kit; QNN, quantum neural network; WDBC, Wisconsin Diagnostic Breast Cancer; LSVC, linear support vector classifier; XGB, extreme gradient boosting; ANN, artificial neural network; VQC, variational quantum classifier; ReLU, Rectified Linear Unit; SHAP, Shapley additive explanations.

Author Contributions: *The corresponding author takes full responsibility that all authors on this publication have met the following required criteria of eligibility for authorship: (a) significant contributions to the conception and design, acquisition of data, or analysis and interpretation of data; (b) drafting or revising the article for intellectual content; (c) final approval of the published article; and (d) agreement to be accountable for all aspects of the article thus ensuring that questions related to the accuracy or integrity of any part of the article are appropriately investigated and resolved. Nobody who qualifies for authorship has been omitted from the list.*

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References

1. Bunch DR, Durant TJ, Rudolf JW. Artificial intelligence applications in clinical chemistry. Clin Lab Med 2023;43:47–69.
2. Herman DS, Rhoads DD, Schulz WL, Durant TJS. Artificial intelligence and mapping a new direction in laboratory medicine: a review. Clin Chem 2021;67:1466–82.
3. Ghazi L, Farhat K, Hoenig MP, Durant TJS, El-Khoury JM. Biomarkers vs machines: the race to predict acute kidney injury. Clin Chem 2024;70:805–19.
4. Smith KP, Wang H, Durant TJS, Mathison BA, Sharp SE, Kirby JE, et al. Applications of artificial intelligence in clinical microbiology diagnostic testing. Clin Microbiol Newsl 2020;42:61–70.
5. Master SR, Badrick TC, Bietenbeck A, Haymond S. Machine learning in laboratory medicine: recommendations of the IFCC working group. Clin Chem 2023;69:690–8.
6. Haymond S, Master SR. How can we ensure reproducibility and clinical translation of machine learning applications in laboratory medicine? Clin Chem 2022;68:392–5.
7. Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. Science 2019;366:447–53.
8. Durant TJS, Knight E, Nelson B, Dudgeon S, Lee SJ, Walliman D, et al. A primer for quantum computing and its applications to healthcare and biomedical research. J Am Med Inform Assoc 2024;31:1774–84.
9. Nielsen MA, Chuang IL. Quantum computation and quantum information. Cambridge (United Kingdom): Cambridge University Press; 2010.
10. Emani PS, Warrell J, Anticevic A, Bekiranov S, Gandal M, McConnell MJ, et al. Quantum computing at the frontiers of biological sciences. Nat Methods 2021;18:701–9.
11. Lau B, Emani PS, Chapman J, Yao L, Lam T, Merrill P, et al. Insights from incorporating quantum computing into drug design workflows. Bioinformatics 2023;39:btac789.
12. Li RY, Di Felice R, Rohs R, Lidar DA. Quantum annealing versus classical machine learning applied to a simplified computational biology problem. npj Quantum Inf 2018;4:14.
13. Phillipson F. Quantum machine learning: benefits and practical examples. QANSWER 2020;2561:51–6.
14. Abdulsalam G, Meshoul S, Shaiba H. Explainable heart disease prediction using ensemble-quantum machine learning approach. Intell Autom Soft Comput 2023;36:761–79.
15. Rani S, Kumar Pareek P, Kaur J, Chauhan M, Bhambri P. Quantum machine learning in healthcare: developments and challenges. In: 2023 IEEE International Conference on Integrated Circuits and Communication Systems (ICICACS) [Internet]. Raichur, India: Institute of Electrical and Electronics Engineers; 2023. p. 1–7. https://ieeexplore.ieee.org/abstract/document/10100075 (Accessed November 2025).
16. Brooks M. Towards quantum machine learning. [Epub] Nature May 24, 2023, as doi: 10.1038/d41586-023-01718-2.
17. Houssein EH, Abohashima Z, Elhoseny M, Mohamed WM. Machine learning in the quantum realm: the state-of-the-art, challenges, and future vision. Expert Syst Appl 2022;194:116512.
18. Flöther FF. The state of quantum computing applications in health and medicine [Internet]. Preprint at http://arxiv.org/abs/2301.09106 (2023).
19. Liang Z, Wang Z, Yang J, Yang L, Shi Y, Jiang W. Can noise on qubits be learned in quantum neural network? A case study on QuantumFlow (invited paper). In: 2021 IEEE/ACM International Conference on Computer Aided Design (ICCAD) [Internet]. Munich, Germany: Institute of Electrical and Electronics Engineers and Association for Computing Machinery; 2021. p. 1–7. https://ieeexplore.ieee.org/abstract/document/9643470 (Accessed September 2024).
20. Pérez-Salinas A, Cervera-Lierta A, Gil-Fuster E, Latorre JI. Data re-uploading for a universal quantum classifier. Quantum 2020;4:226.
21. Pérez-Salinas A, López-Núñez D, García-Sáez A, Forn-Díaz P, Latorre JI. One qubit as a universal approximant. Phys Rev A 2021;104:012405.
22. Moussa C, Patel YJ, Dunjko V, Bäck T, van Rijn JN. Hyperparameter importance and optimization of quantum neural networks across small datasets. Mach Learn 2024;113:1941–66.
23. Aminpour S, Banad Y, Sharif S. Strategic Data Re-Uploads: A Pathway to Improved Quantum Classification Data Re- Uploading Strategies for Improved Quantum Classifier Performance. Preprint at https://doi.org/10.48550/arXiv.2405.09377 (2024).
24. Wach NL, Rudolph MS, Jendrzejewski F, Schmitt S. Data re-uploading with a single qudit. Preprint at https://doi.org/10.48550/arXiv.2302.13932 (2024).
25. Wilkes EH, Emmett E, Beltran L, Woodward GM, Carling RS. A machine learning approach for the automated

interpretation of plasma amino acid profiles. Clin Chem 2020;66:1210–8.
26. Lee ES, Durant TJS. Supervised machine learning in the mass spectrometry laboratory: a tutorial. J Mass Spectrom Adv Clin Lab 2022;23:1–6.
27. Fisher RA. The use of multiple measurements in taxonomic problems. Ann Eugen 1936;7:179–88.
28. Wolberg WH, Street WN, Mangasarian OL. Machine learning techniques to diagnose breast cancer from image-processed nuclear features of fine needle aspirates. Cancer Lett 1994;77:163–71.
29. Lee SJ, Durant TJS, Dudgeon S, Nelson B, Young P, Horn G, et al. Quantum Neural Network Tuning and Performance Evaluation for a Breast Cancer Dataset. Preprint at https://doi.org/10.1101/2025.10.03.25336905 (2025).
30. Kelly M, Longjohn R, Nottingham K. The UCI Machine Learning Repository [Internet]. https://archive.ics.uci.edu/ (Accessed September 2024).
31. Hur T, Kim L, Park DK. Quantum convolutional neural network for classical data classification. Quantum Mach Intell 2022;4:3.
32. Rathore AS, Kumar N, Choudhury S, Mehta NK, Raghava GPS. Prediction of hemolytic peptides and their hemolytic concentration. Commun Biol 2025;8:1–14.
33. Ahmed S. Data-reuploading classifier. PennyLane Demos [Internet]. Xanadu; 2019. https://pennylane.ai/qml/demos/tutorial_data_reuploading_classifier/ (Accessed May 2024).
34. Havlíček V, Córcoles AD, Temme K, Harrow AW, Kandala A, Chow JM, et al. Supervised learning with quantum-enhanced feature spaces. Nature 2019;567:209–12.
35. Ranga D, Rana A, Prajapat S, Kumar P, Kumar K, Vasilakos AV. Quantum machine learning: exploring the role of data encoding techniques, challenges, and future directions. Mathematics 2024;12:3318.
36. Harrison JH Jr, Gilbertson JR, Hanna MG, Olson NH, Seheult JN, Sorace JM, et al. Introduction to artificial intelligence and machine learning for pathology. Arch Pathol Lab Med 2021;145:1228–54.
37. Çubukçu HC, Topcu Dİ, Yenice S. Machine learning-based clinical decision support using laboratory data. Clin Chem Lab Med 2024;62:793–823.
38. Pfob A, Lu S-C, Sidey-Gibbons C. Machine learning in medicine: a practical introduction to techniques for data pre-processing, hyperparameter tuning, and model comparison. BMC Med Res Methodol 2022;22:282.
39. Biamonte J, Wittek P, Pancotti N, Rebentrost P, Wiebe N, Lloyd S. Quantum machine learning. Nature 2017;549:195–202.